

SOFTWARE DEVELOPMENT INTERNSHIP PROGRAM

AI-ASSISTED WEB DEVELOPMENT & WORKFLOW AUTOMATION

AN INTERNSHIP PROJECT REPORT

Submitted by

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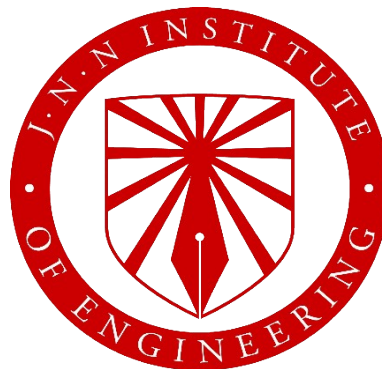
in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science



J.N.N Institute of Engineering

(An Autonomous Institute , Affiliated to Anna University, Chennai)

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S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.



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PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team



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PROGRAM SPECIFIC OBJECTIVES (PSOs)

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PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight , hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.



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BONAFIDE CERTIFICATE

Certified that this internship project report
“AI-ASSISTED WEB DEVELOPMENT & WORKFLOW AUTOMATION”
(Software Development Internship) is the bonafide work of “SAKTHIVEL B”
(Reg No: 110725102212) who carried out the project under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan

Professor and Head,
Department of Artificial Intelligence
& Data Science Engineering,
**J.N.N Institute of Engineering,
Chennai**

Signature of the Supervisor with date

Miss. Brindha . S

Project Supervisor,
Department of Artificial Intelligence
& Data Science Engineering,
**J.N.N Institute of Engineering,
Chennai**

Certified that the above students have attend of viva voce during the exam
held on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, we express our sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.**, and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYAHANDRAN M.S.**, for providing necessary facilities for the completion of our project.

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We take this opportunity to express our gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving us continuous encouragement for completing our project.

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Finally, we would like to thank our parents and friends for their constant support and encouragement.

ABSTRACT

Software development is changing quickly with the arrival of AI-assisted coding tools, and companies now look for developers who can combine strong fundamentals with these tools. This report documents the work carried out during the Software Developer Internship at QIQ AI, Chennai, completed between 10th May 2026 and 31st May 2026 under the guidance of the mentor, Moghilan.

The internship began from the basics of web coding and moved through nine connected activities. The intern learned HTML, CSS and JavaScript; used Google Antigravity to build three websites – Coop AI, a home-services platform inspired by Urban Company, Buhari, a restaurant website, and a Snapchat-style landing page; practised Python basics; learned Git and GitHub; and built an n8n workflow that sends payment-due notifications automatically through Gmail. The basics of digital marketing were also studied.

The work shows that AI-assisted development can greatly reduce the time needed to produce a first working version of a website, but that sound fundamentals, careful code review and testing remain essential. The internship gave the intern a portfolio of three responsive websites, a working automation workflow and practical exposure to a professional software environment.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Software is now part of almost every business, from a neighbourhood restaurant that takes orders online to a home-service platform that books a technician within an hour. Behind each of these services is a website or an application that someone had to design, code, test and publish. Traditionally this work needed long hours of manual coding in HTML, CSS and JavaScript. Today, AI-assisted development tools can read a written instruction, plan the work and generate working code, so that a developer spends more time on design, review and correction than on typing.

This report presents the work carried out during the **Software Developer Internship** at **QIQ AI, Chennai**, from 10th May 2026 to 31st May 2026. The internship was held offline at the company, under the guidance of the mentor, Moghilan. The intern joined as a fresher, and the training therefore began from the basics: what web coding is, how a website is built, how an AI tool can be used to build one, how Python programs are written, how simple code automation works and how products are marketed online.

During the internship three websites were built using **Google Antigravity**, an AI-assisted coding environment: **Coop AI**, a home-services booking website inspired by Urban Company; **Buhari**, a restaurant website for Buhari Restaurant; and a **Snapchat-style landing page**. The intern also learned Git and GitHub for version control, practised Python basics, and built a workflow in **n8n** that automatically sends a payment-due notification to a customer by Gmail.

1.2 Internship Details

Table 1.1 – Details of the internship

Particulars	Details
Name of the Intern	SAKTHIVEL B
Register Number	110725102212
Department	Artificial Intelligence & Data Science
Year / Semester	II Year / III Semester (Batch 2025 – 2029)
Institution	J.N.N Institute of Engineering, Chennai
Organization	QIQ AI, No. 2/41, Masilamani Road, Balaji Nagar, Royapettah, Chennai – 600 041
Role	Software Developer Intern
Mode of Internship	Offline (at the company)
Mentor / Trainer	Moghilan
Duration	22 days (10.05.2026 to 31.05.2026)
Certificate issued by	Mr. Abinav Satish, Founder, QIQ AI (dated 18.08.2026)

The internship did not follow a single fixed project title. It was a guided training programme in which the intern learned the fundamentals first and then applied them to practical builds. The

completion certificate issued by QIQ AI, which confirms the role and the period of the internship, is attached in the Appendix.

1.3 Objectives

- To understand the fundamentals of web coding and how a website is structured using HTML, CSS and JavaScript.
- To learn how to build complete, responsive websites with the help of an AI-assisted coding tool (Google Antigravity).
- To develop real-world websites – a home-services platform, a restaurant website and a social-media landing page – and to review and correct the generated code.
- To learn the basics of Python programming through small practical programs.
- To learn the basics of code automation by building a no-code workflow in n8n that sends payment-due notifications through Gmail.
- To learn version control using Git and GitHub, and the basics of digital marketing for promoting a website or a product.

1.4 Overview of the Internship Activities

The internship was divided into the following connected activities, each building on the previous one:

- **Module 1 – Web Coding Fundamentals:** how a web page works; HTML structure, CSS styling and basic JavaScript behaviour.
- **Module 2 – AI-Assisted Development:** setting up Google Antigravity, writing prompts, reviewing and improving the generated code.
- **Module 3 – Coop AI Website:** a home-services booking website similar in idea to Urban Company.
- **Module 4 – Buhari Restaurant Website:** a restaurant website with a home page, menu, about and contact sections.
- **Module 5 – Snapchat-Style Landing Page:** a bold, colourful landing page with features and download buttons.
- **Module 6 – Python Basics:** variables, lists, loops, conditions, functions and dictionaries.
- **Module 7 – Git and GitHub:** repositories, commits and pushing a project to GitHub.
- **Module 8 – Code Automation with n8n:** an automatic payment-due reminder sent by Gmail.
- **Module 9 – Digital Marketing Basics:** online promotion, audience, content and search visibility.



Fig. 1.1 – Phases of the internship work flow

1.5 Problem Statement

Small businesses and start-ups need a working website quickly, but hiring a full development team is expensive and slow. At the same time, a beginner who tries to learn web development from books alone often takes weeks before producing a single usable page. Repetitive office tasks, such as sending a payment reminder to every customer, are also still done by hand, which wastes time and leads to missed or late messages.

The internship therefore addressed a practical question: how can a beginner use modern AI-assisted tools, together with a sound understanding of the fundamentals, to build professional-looking websites quickly, and how can simple automation remove repetitive manual work?

1.6 Scope of the Work

The work was carried out as a learning-oriented internship and is limited to the following scope.

- **Included:** front-end development of three websites using HTML, CSS and JavaScript with AI assistance;
- **Included:** basic Python programs, Git and GitHub usage, and one n8n automation workflow;
- **Included:** introduction to digital marketing concepts for promoting a website;
- **Not included:** a production back-end, a payment gateway, a live database and cloud deployment with a custom domain, which are suggested as future work in Chapter 6.

1.7 Organisation of the Report

The report is organised into six chapters. Each chapter builds on the one before it, so that the reader can move from the purpose of the internship to its methods, its execution and finally its outcomes.

- **Chapter 1 – Introduction:** states the purpose of the internship, the problem, the objectives, the scope and the activities.
- **Chapter 2 – Literature Review:** explains web development, AI-assisted coding, Python, automation, version control and digital marketing.
- **Chapter 3 – Methodology:** describes the approach, the schedule, the AI-assisted workflow, the requirements and the testing method.
- **Chapter 4 – Implementation:** presents each module with code, screenshots, tests and the difficulties faced.
- **Chapter 5 – Results and Discussion:** discusses the websites built, the skills gained and the comparison with conventional coding.
- **Chapter 6 – Conclusion and Future Work:** summarises the work, the skills, the limitations and the future scope.

CHAPTER 2

LITERATURE REVIEW

This chapter gives the background of the concepts, techniques and tools used during the internship.

2.1 Web Development

A website is a collection of web pages that a browser downloads and displays. Every page is built from three core technologies. **HTML** (HyperText Markup Language) defines the structure and content – headings, paragraphs, images, links and forms. **CSS** (Cascading Style Sheets) controls the appearance – colours, fonts, spacing and layout. **JavaScript** adds behaviour – menus that open, forms that are checked and content that changes without reloading the page [1], [2].

Modern websites must work on screens of every size. Responsive design achieves this with flexible layouts, relative units and media queries, so that the same page rearranges itself for a phone, a tablet and a desktop. Front-end development concentrates on what the visitor sees and uses, while back-end development deals with servers and databases; the websites built in this internship are front-end projects.

2.2 AI-Assisted Software Development

AI-assisted development uses large language models to read a description written in ordinary language and to produce, explain or modify source code. Early tools only completed the line a developer was typing. Newer agentic tools plan a whole task, create several files, run the program and check the result in a browser. Google Antigravity, released in November 2025, is an example: it is built on a Visual Studio Code style editor and adds an agent panel that can plan, write and verify code across the editor, terminal and a built-in browser [3], [4].

Antigravity reports its progress through artifacts such as task lists, implementation plans and screenshots, which makes it easier for the developer to review what the agent has done instead of trusting it blindly. This idea of human review is important: the developer remains responsible for the quality of the final code.

2.3 Generative AI in Coding

The quality of AI-generated code depends strongly on the quality of the instruction, called the prompt. A good prompt states the goal, the pages required, the style, the technologies and any constraints. Prompt engineering is therefore the skill of writing instructions that give the model enough context to produce useful output. Even with a good prompt, generated code can contain mistakes, unused code or design choices that do not match the requirement, so it must always be read, tested and corrected by the developer.

- **Speed:** a working first version of a page can be produced in minutes.
- **Learning aid:** the generated code shows how a layout or feature can be written, which helps a beginner learn faster.
- **Risk:** the output may look correct but fail on small screens or contain hidden errors, so

testing cannot be skipped.

2.4 Python Programming

Python is a high-level, general-purpose language known for its readable syntax. A beginner learns variables, data types, operators, conditions (if-else), loops (for and while), lists, dictionaries and functions, and can already write useful programs such as a mark calculator or a bill generator [5]. Python is also widely used for automation, because a few lines can read files, send messages or call web services. This is why it is taught in the internship alongside web development.

2.5 Code Automation and Workflow Tools

Automation means making a computer carry out a repeated task without manual effort. Workflow-automation tools such as n8n let the user connect applications visually. A workflow starts with a trigger (for example, a daily schedule), passes data through nodes (read a spreadsheet, check a condition) and ends with an action (send an e-mail) [6]. Because each step is a node on a canvas, even a beginner can understand and change the flow. Gmail is commonly used as the action node to send notifications to customers [7].

2.6 Version Control with Git and GitHub

Git is a distributed version-control system that records every change to a project as a commit, so that earlier versions can always be recovered [8]. GitHub is an online service that hosts Git repositories and allows a project to be backed up, shared and developed by several people. Using Git is a standard professional practice and was introduced to the intern during the internship [9].

2.7 Digital Marketing

Building a website is only the first step; people must also find it. Digital marketing is the promotion of a product or service through online channels such as search engines, social media, e-mail and content. Important ideas include identifying the target audience, writing clear and attractive content, using a strong call to action and measuring the response [10]. For a restaurant such as Buhari, this may mean showing the menu clearly, adding a reservation button and sharing photographs of dishes on social media.

2.8 Tools and Their Purpose

Table 2.1 – Tools and technologies used

Tool / Technology	Purpose in the internship
HTML5	Structure and content of every web page
CSS3	Layout, colours, typography and responsive design
JavaScript	Interactive behaviour such as menus and form handling
Google Antigravity	AI-assisted coding environment used to plan and generate the websites
VS Code / Editor	Reading, editing and debugging the generated code
Web browser (Chrome)	Previewing pages and testing responsiveness

Tool / Technology	Purpose in the internship
Python 3	Learning programming basics and writing small programs
Git and GitHub	Version control and hosting of project files
n8n	Visual workflow automation for the payment-due notification
Gmail	Sending the automatic notification e-mail to the customer

2.9 Existing Approaches and Their Limitations

Websites are usually built in one of three ways. The first is hand-coding every page, which gives full control but is slow for a beginner. The second is a website builder with ready-made templates, which is fast but limits the design and does not teach the developer how the site works. The third is hiring an agency, which is costly for a small business.

- **Manual coding:** time-consuming and demands strong knowledge of syntax from the start.
- **Template builders:** quick, but results look similar and customisation is limited.
- **Agency development:** professional, but expensive and slow for small projects.
- **Manual office work:** reminders and notifications sent by hand are slow and easy to forget.

2.10 Need for the Approach Followed

The approach followed in the internship combines the strengths of these methods. The intern first learned the fundamentals, so that the generated code could be understood and corrected. An AI-assisted tool then removed the repetitive typing and produced a first version quickly. Finally, Git preserved the work and n8n automated a repetitive business task. Together they show how a single beginner can deliver working websites and automation within a few weeks.

CHAPTER 3 METHODOLOGY

3.1 Overall Approach

The internship followed a step-by-step approach in which each stage prepared the intern for the next. The fundamentals of web coding were learned first, the AI-assisted tool was then introduced, three websites were planned and built, and the supporting skills of Python, Git, automation and marketing were added. Every website was previewed in the browser and corrected before it was considered complete.

3.2 Internship Schedule

The internship ran for three weeks at the QIQ AI office. Table 3.1 summarises the topics covered in each week.

Table 3.1 – Week-wise activities during the internship

Period	Activities
Week 1 10 – 16 May	Introduction to web coding; HTML structure; CSS styling; basic JavaScript; setting up the tools.
Week 2 17 – 23 May	AI-assisted development with Google Antigravity; prompt writing; building the Coop AI and Buhari websites.
Week 3 24 – 31 May	Snapchat-style landing page; Python basics; Git and GitHub; n8n automation; digital marketing basics; final review.

3.3 AI-Assisted Coding Workflow

For every website the same repeatable workflow was followed. The idea and the required pages were written down, a clear prompt was given to the AI agent, the generated files were read and edited, and the result was tested in the browser. If a problem was found, a corrected prompt or a manual edit was used and the page was tested again.



Fig. 3.1 – AI-assisted website development workflow

3.4 Website Development Methodology

- **Requirement:** decide the purpose of the site, the target visitor and the pages needed.
- **Design:** choose a colour theme, fonts and layout for each page.
- **Prompt:** describe the website to the AI agent with the pages, sections and style.
- **Review:** read the generated HTML, CSS and JavaScript and remove or correct what is not needed.
- **Test:** open every page in the browser, check links, buttons and screen sizes.
- **Save:** commit the project to Git and push it to GitHub.

3.5 Python Learning Methodology

Python was learned through short, practical programs rather than long theory. Each concept – variables, input and output, lists, loops, conditions, functions and dictionaries – was practised with a small example and its output was checked immediately. The examples are given in Chapter 4.

3.6 Automation Workflow Methodology

The automation task was to remind customers of an upcoming payment without any manual work. The workflow was designed first on paper as a sequence of five steps and then built node by node in n8n.



Fig. 3.2 – Steps of the payment-due notification workflow

3.7 System Requirements

The work does not need special equipment. Table 3.2 lists the hardware and software that were used.

Table 3.2 – Minimum hardware and software requirements

Item	Requirement
Processor	Intel Core i3 or equivalent, or higher
Memory (RAM)	8 GB recommended for the AI-assisted editor
Storage	About 2 GB of free disk space
Operating system	Windows 10 / 11, Linux or macOS
Browser	Google Chrome (latest version)
Editor	Google Antigravity / VS Code
Language	Python 3, JavaScript
Other software	Git, GitHub account, n8n, Gmail account
Internet	Required for the AI agent, GitHub and n8n

3.8 Functional and Non-Functional Requirements

Functional requirements

- Each website shall have a navigation bar, a home page and clearly separated sections.
- The Coop AI site shall list services and provide a booking form.
- The Buhari site shall show the menu with names, descriptions and prices.
- The automation shall send an e-mail to the customer when a payment is about to fall due.
- The project files shall be stored in a Git repository.

Non-functional requirements

- **Responsiveness:** the pages should adjust to phone, tablet and desktop screens.
- **Usability:** menus, buttons and text should be easy to read and use.

- **Maintainability:** the code should be organised in separate HTML, CSS and JavaScript files.
- **Reliability:** the automation should run on its schedule without manual action.

3.9 Testing and Debugging Method

Testing was done manually in the browser. Each page was opened, every link and button was clicked, and the window was resized to check the layout. Errors were found by reading the browser developer console and by comparing the page with the intended design. Errors were then fixed by editing the code or by giving the AI agent a more specific instruction.

CHAPTER 4

IMPLEMENTATION

This chapter describes the practical execution of each module, including the code and the screenshots taken from the work environment.

4.1 Module 1: Web Coding Fundamentals

Services used: HTML5, CSS3, JavaScript, Google Chrome

The internship began with the basics. The intern learned how the browser reads an HTML file, how tags are nested, and how a separate CSS file styles them. A first page with a heading, a paragraph and a button was written by hand to understand the structure before any AI tool was used.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Page</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first web page.</p>
  <button onclick="greet()">Click me</button>
  <script src="script.js"></script>
</body>
</html>
```

Code 4.1 – Structure of a basic HTML page

```
body {
  font-family: Arial, sans-serif;
  background-color: #f7f6ff;
  text-align: center;
}
button {
  background: #5b3df5;
  color: #ffffff;
  padding: 10px 24px;
  border: none;
  border-radius: 8px;
}
/* script.js */
function greet() { alert("Welcome to my website!"); }
```

Code 4.2 – Basic CSS styling and a JavaScript function

4.2 Module 2: AI-Assisted Development with Google Antigravity

Services used: Google Antigravity, Gemini model, Chrome

After the fundamentals, Google Antigravity was introduced. Its editor looks like VS Code, with the project files on the left, the code in the centre and the AI agent panel on the right. The intern wrote a prompt describing the website; the agent produced a plan, created the HTML, CSS and JavaScript files and could preview the result. The intern then read the code, removed unnecessary parts and changed colours, text and layout to match the requirement.

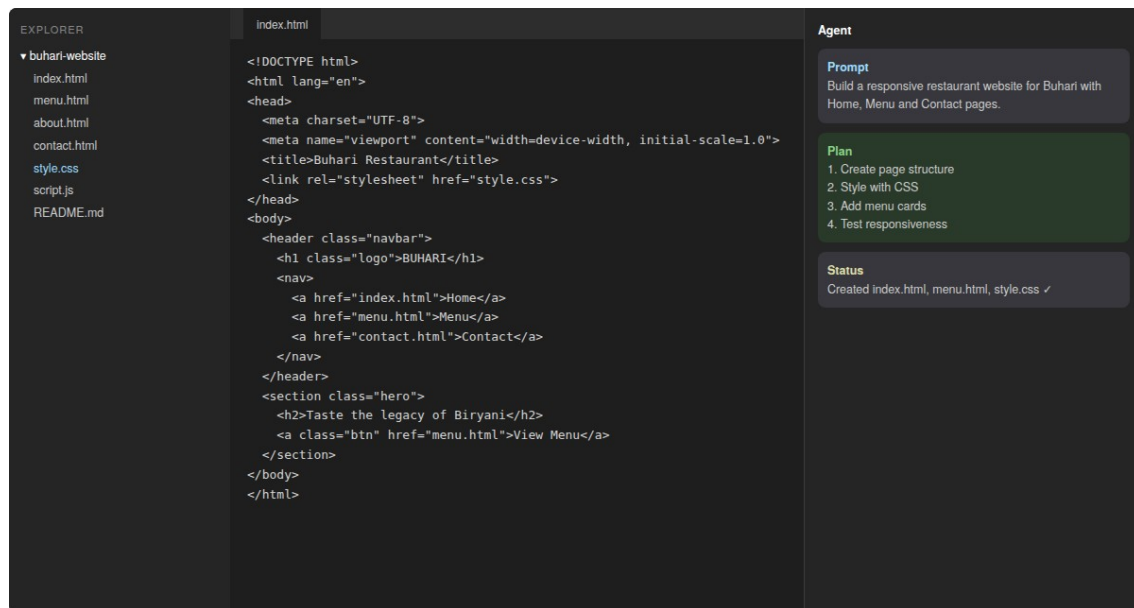


Fig. 4.1 – AI-assisted development of a website in the coding environment

Example prompt used

```
Build a responsive restaurant website for "Buhari Restaurant".
Pages: Home, Menu, About, Contact.
Style: dark maroon and gold theme, large hero section,
menu cards with price and an "Order" button,
and a "Reserve a Table" button in the navigation bar.
```

Code 4.3 – Prompt given to the AI agent

4.3 Module 3: Coop AI – Home Services Website

Services used: HTML, CSS, JavaScript, Google Antigravity

Coop AI is a home-services booking website similar in idea to Urban Company. A visitor can search for a service such as home cleaning, salon at home, AC repair, electrician or plumbing, read how the process works and book a service by filling in a short form. The home page has a navigation bar with Login and Sign up buttons, a hero section with a search box, a grid of service tiles and a three-step “How it works” section.

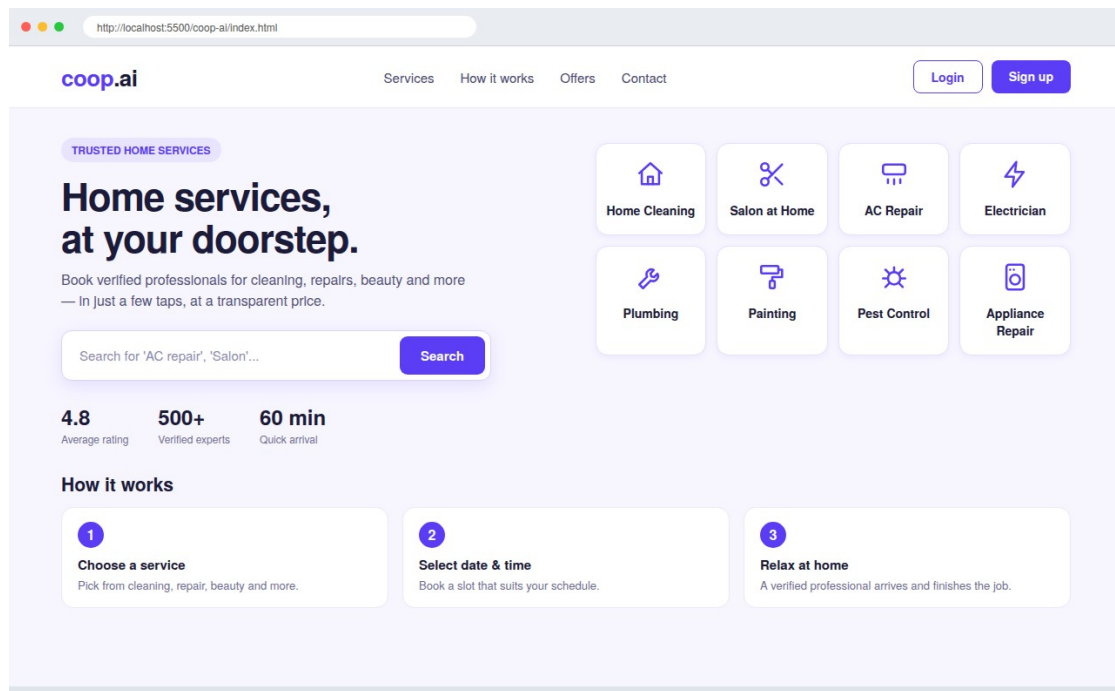


Fig. 4.2 – Coop AI home page

The services page lists each service with its rating and starting price, and a booking panel collects the customer's name, phone number, service, preferred date and time, and address. The pages use a purple theme with rounded cards to give a modern, trustworthy appearance.

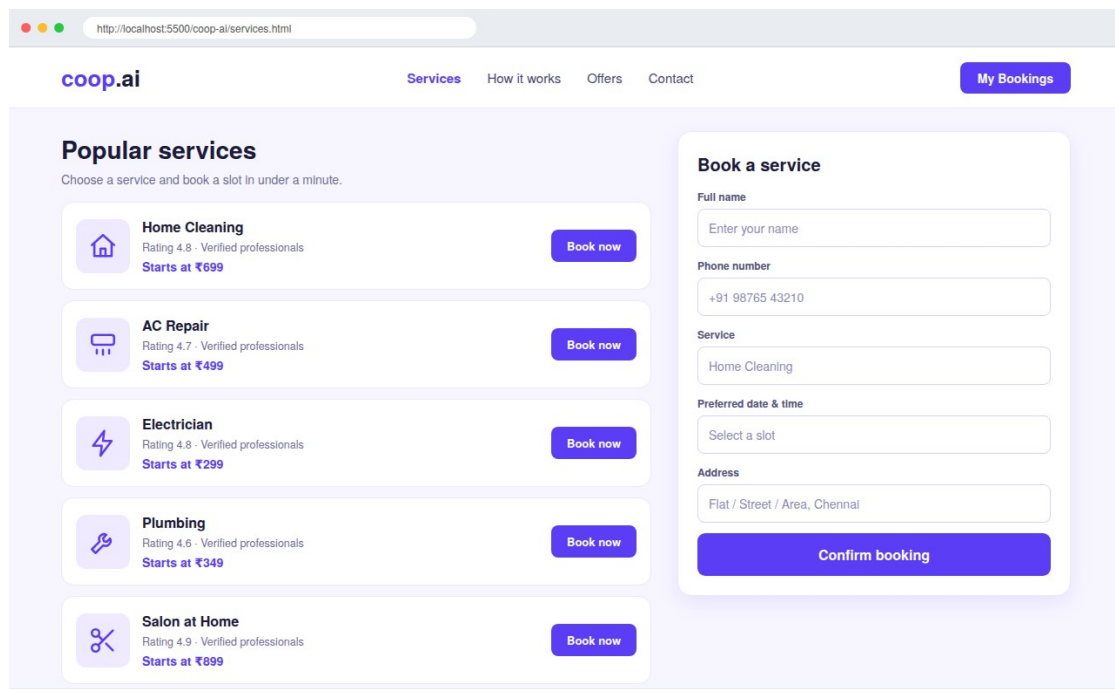


Fig. 4.3 – Coop AI services and booking page

4.4 Module 4: Buhari Restaurant Website

Services used: HTML, CSS, JavaScript, Google Antigravity

The Buhari website was designed for Buhari Restaurant. A dark maroon and gold colour theme was chosen to suit a traditional biryani restaurant. The home page has a hero section with the restaurant

slogan, “View Menu” and call buttons, and a “Today’s Specials” row of cards showing dishes such as Chicken 65, Mutton Biryani, Butter Naan and Paneer Tikka with their prices.

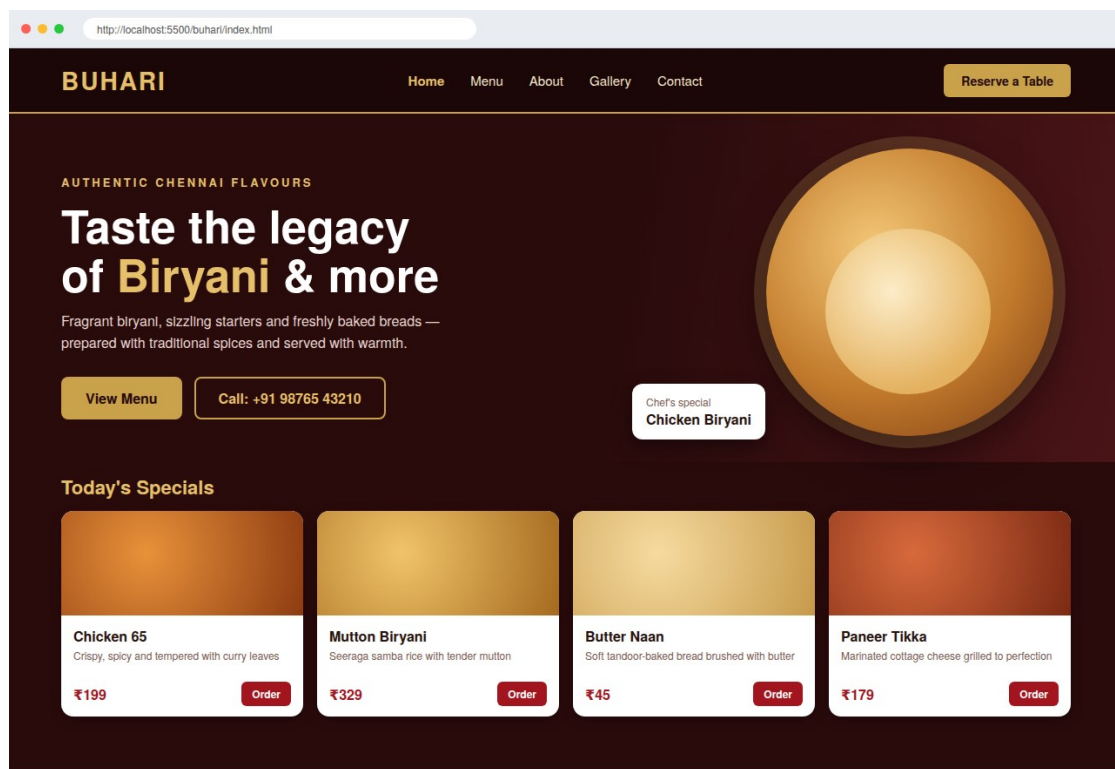


Fig. 4.4 – Buhari Restaurant home page

The menu page groups the dishes into Starters, Biryani, and Breads and Desserts, each with a short description and a price, together with category buttons at the top. The “Reserve a Table” button remains visible in the navigation bar on every page so that a visitor can always contact the restaurant.

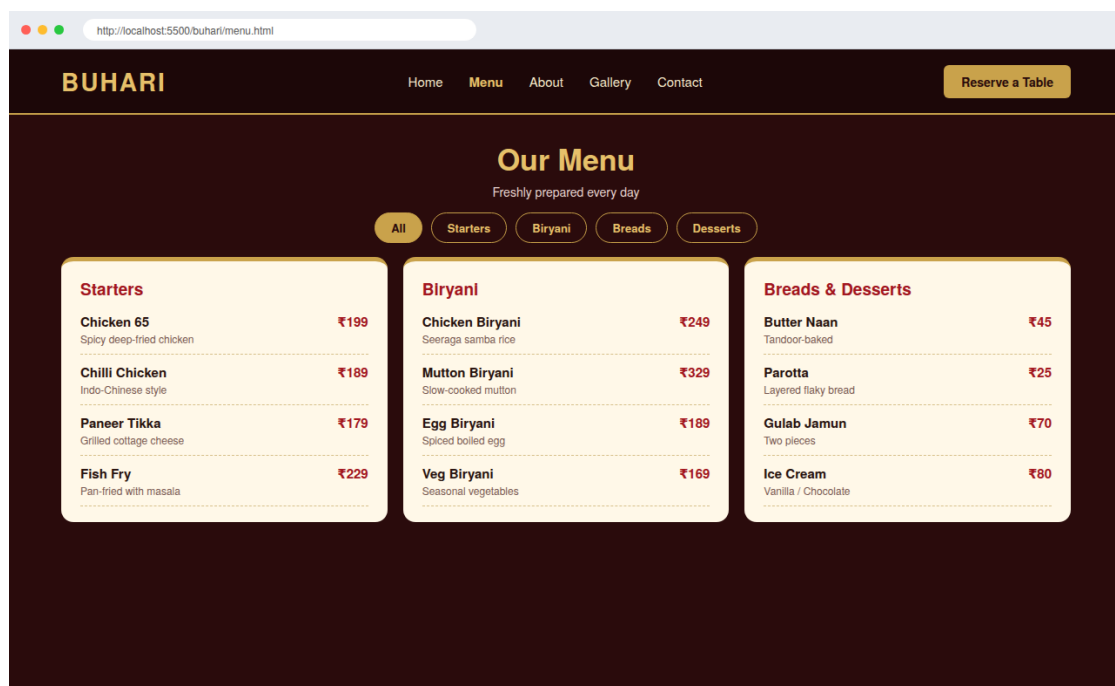


Fig. 4.5 – Buhari Restaurant menu page

4.5 Module 5: Snapchat-Style Landing Page

Services used: HTML, CSS, Google Antigravity

As a third exercise, a landing page in the style of the Snapchat website was built to practise a different design approach: a bright yellow background, very large bold headlines, a phone mock-up and four feature cards for Chat, Stories, Spotlight and Lenses. This project is an educational practice page and is not connected to the real Snapchat service. It helped the intern understand how large brands present a single clear message with strong colour and typography.

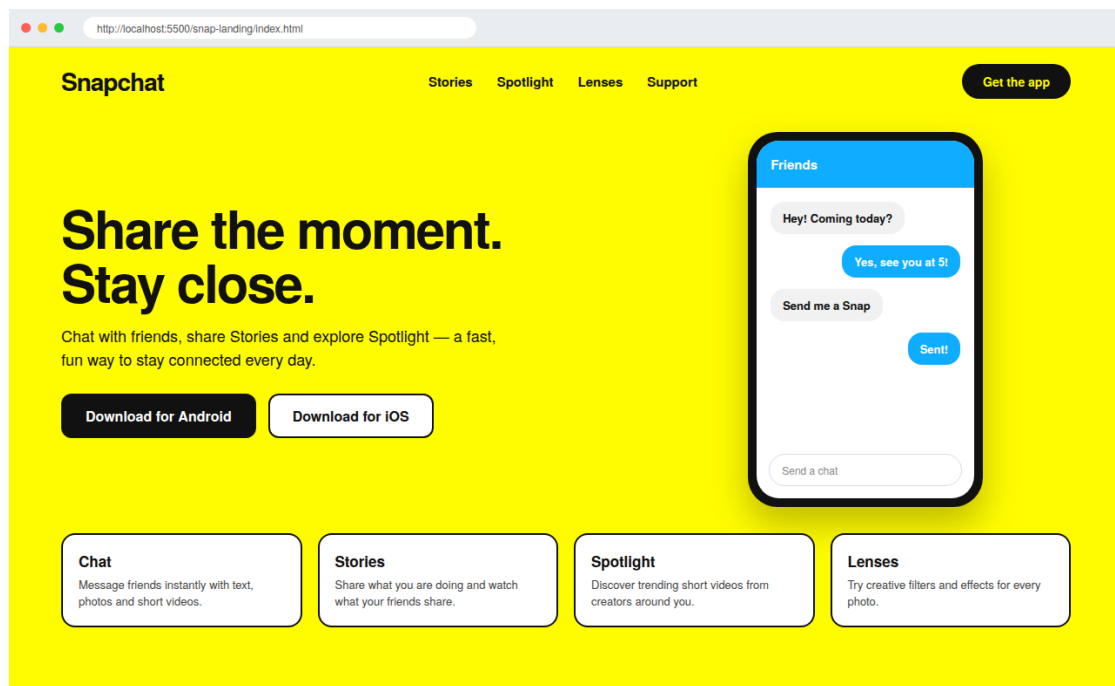


Fig. 4.6 – Snapchat-style landing page (practice project)

4.6 Module 6: Python Basics

Services used: Python 3, VS Code, Terminal

The Python basics were practised with small programs. Code 4.4 shows a program that uses variables, a list, a loop, a condition, a function and a dictionary to calculate a total, an average, a pass/fail result and a simple restaurant bill. The output of the program is shown in Fig. 4.7.

```

name = "Sakthivel"
age = 19
print("Hello,", name)
print("Next year age:", age + 1)

marks = [78, 85, 92, 67, 88]
total = sum(marks)
print("Total:", total, "Average:", total / len(marks))

for m in marks:
    print(m, "-> Pass" if m >= 50 else "-> Fail")

def square(n):
    return n * n
print("Square of 7:", square(7))

menu = {"Chicken Biryani": 249, "Butter Naan": 45}
bill = 0
for item, price in menu.items():
    bill += price
print("Bill amount: Rs.", bill)

```

Code 4.4 – Python basics: variables, lists, loops, functions and dictionaries



```

Terminal - python basics.py

$ python basics.py
Hello, Sakthivel
Next year age: 20
Total: 410 Average: 82.0
78 -> Pass
85 -> Pass
92 -> Pass
67 -> Pass
88 -> Pass
Square of 7: 49
Bill amount: Rs. 294

```

Fig. 4.7 – Output of the Python basics program

4.7 Module 7: Git and GitHub

Services used: Git, GitHub

The intern learned to place a project under version control. A repository was initialised with `git init`, the files were added with `git add`, and each meaningful change was saved with `git commit` and a clear message. The project can then be pushed to a remote GitHub repository, which acts as a backup and makes the work easy to share.

```

git init
git add .
git commit -m "Initial commit: Buhari website"
git add menu.html
git commit -m "Add menu page"
git log --oneline
git remote add origin https://github.com/<username>/buhari-website.git
git push -u origin main

```

Code 4.5 – Basic Git and GitHub commands

```

Terminal - Git Bash

$ git status
## main

$ git log --oneline
dd8b5db Add menu page
d82ac97 Initial commit: Buhari website

$ git remote add origin https://github.com/<username>/buhari-website.git
$ git push -u origin main

```

Fig. 4.8 – Git commits recorded for the project

4.8 Module 8: Code Automation with n8n

Services used: n8n, Google Sheets, Gmail

The automation task was to send a payment-due notification to a customer automatically. The workflow in n8n starts with a Schedule Trigger that runs every morning at 9:00 AM. A Google Sheets node reads the list of customers and their due dates, and an IF node checks whether the due date is within the next two days. For the customers who satisfy the condition, a Gmail node sends the reminder and a final node marks the record as “reminder sent”, so that nobody receives the same reminder twice. Customers who are not due are skipped.

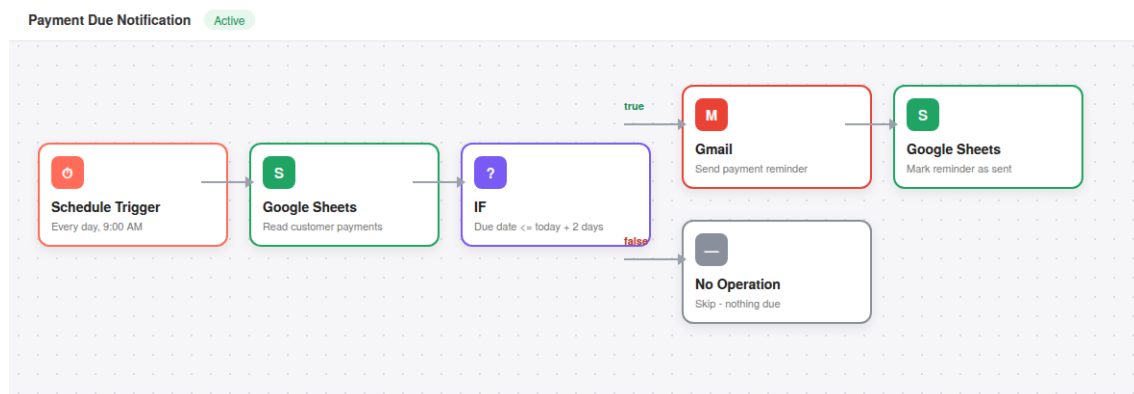


Fig. 4.9 – n8n workflow for the automatic payment-due notification

Notification message template

```

Subject: Payment Reminder: Invoice #{{ $json.invoice }} is due in 2 days

Dear {{ $json.name }},

This is a friendly reminder that your payment of Rs. {{ $json.amount }}
for Invoice #{{ $json.invoice }} is due on {{ $json.due_date }}.
Please complete the payment at your earliest convenience.

Thank you,
Accounts Team

```

Code 4.6 – Message template used in the Gmail node

When the workflow runs, the customer receives an e-mail in the inbox similar to the one shown in Fig. 4.10. Once the workflow is switched on, no manual effort is needed to send the reminders.

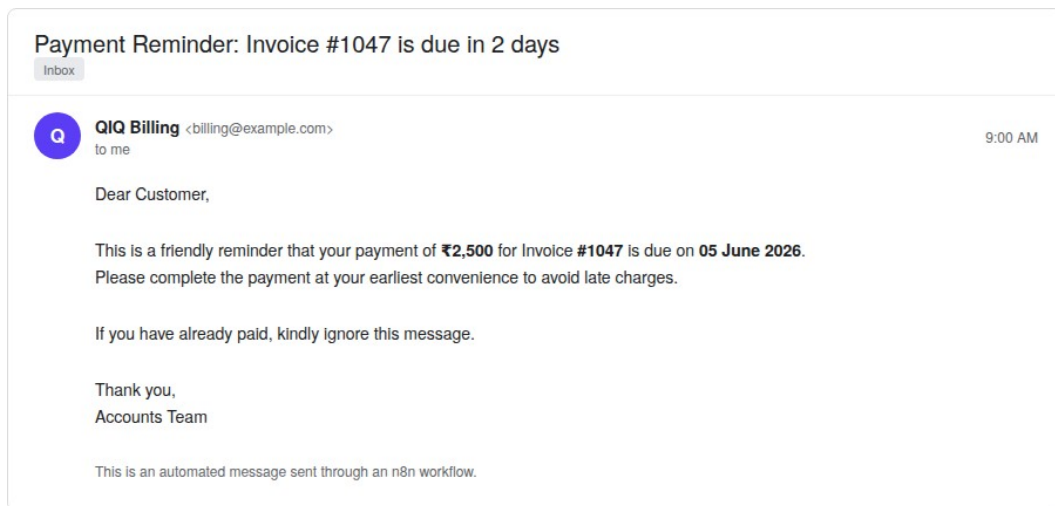


Fig. 4.10 – Payment reminder e-mail received by the customer

4.9 Module 9: Digital Marketing Basics

The training also introduced the basics of marketing a website once it is built. Table 4.1 lists the main ideas and how they were related to the websites developed.

Table 4.1 – Marketing concepts and their use in the projects

Concept	Meaning	Use in the projects
Target audience	The group of people the product is meant for	Home owners for Coop AI; food lovers for Buhari
Clear message	A short headline that explains the offer	“Home services, at your doorstep”
Call to action	A button that tells the visitor what to do next	“Book now”, “Reserve a Table”
Social media	Sharing content on platforms to reach people	Posting dish photographs and offers
Search visibility	Making the site easy to find in search engines	Meaningful titles and headings in the pages

4.10 Testing of the Websites and Automation

Each website and the automation were tested using the checks given in Table 4.2.

Table 4.2 – Test checklist

Area	Check performed	Result
Navigation	Every menu link opens the correct page	Working
Buttons	Search, Book now, View Menu and Reserve a Table respond on click	Working
Responsive layout	Browser window resized to phone, tablet and desktop widths	Layout adjusts correctly
Content	Text, prices and images appear without overlap	Correct
Python programs	Program run in the terminal and output compared with expected values	Output as expected

Area	Check performed	Result
Git	Commits listed with git log	Commits recorded
Automation	Test customer with a near due date receives the e-mail	Notification sent

4.11 Challenges Faced and Solutions

Table 4.3 – Difficulties and how they were solved

Challenge	Solution
Generated layout did not fit small screens	Added a viewport tag and flexible widths and tested at different window sizes
The AI added unused code and extra sections	Read the files, removed unneeded parts and refined the prompt
Colours and fonts looked inconsistent	Defined one colour theme and used it across all pages
Understanding the Git commands	Practised init, add, commit and log on a small project first
The automation sent the same reminder more than once	Added a “reminder sent” update so that a record is not processed again

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents the outcomes of the internship, the skills gained and a comparison between conventional and AI-assisted coding.

5.1 Websites Developed

Table 5.1 – Summary of the websites developed

Website	Purpose	Main sections
Coop AI	Home-services booking (Urban Company style)	Home, Services, How it works, Booking form, Login / Sign up
Buhari	Restaurant website for Buhari Restaurant	Home, Today's Specials, Menu, About, Contact, Reserve a Table
Snapchat-style page	Practice landing page	Hero, phone mock-up, Chat, Stories, Spotlight and Lenses cards

All three websites were completed with a consistent theme, a clear navigation bar and responsive layouts. Each one uses a different colour identity – purple for Coop AI, maroon and gold for Buhari and yellow and black for the landing page – which shows how the same technologies can produce very different looks.

5.2 Features Implemented

- Navigation bars with working links and call-to-action buttons.
- Hero sections with clear headlines and search or action buttons.
- Service tiles, menu cards and feature cards built with CSS grid and flexbox.
- A booking form for the Coop AI services page.
- Category-wise menu with prices for the Buhari website.
- An automatic Gmail payment reminder with a duplicate-prevention step.

5.3 Skills Acquired

Table 5.2 – Skills gained during the internship

Area	Skills gained
Web development	HTML structure, CSS styling, JavaScript basics and responsive design
AI-assisted coding	Prompt writing, agent-based development, reviewing generated code
Python	Variables, lists, loops, conditions, functions and dictionaries
Version control	Git commands and pushing a project to GitHub
Automation	Building an n8n workflow with a trigger, a condition and a Gmail action
Marketing	Audience, message, call to action and online promotion

Area	Skills gained
Professional skills	Working in an office, following a mentor's guidance and documenting the work

5.4 Comparison of Conventional and AI-Assisted Coding

Table 5.3 – Conventional coding compared with AI-assisted coding

Aspect	Conventional coding	AI-assisted coding
First version of a page	Written line by line by the developer	Generated from a prompt, then reviewed
Speed	Slower for a beginner	Much faster for the first draft
Learning value	Deep understanding of each line	Good when the code is read and understood
Errors	Mostly typing and logic mistakes	Hidden design or logic mistakes that need review
Developer's role	Writes every line	Plans, instructs, reviews and corrects
Best use	Learning the fundamentals	Producing and improving real projects quickly

The comparison shows that the two approaches are complementary. The fundamentals learned in the first week were necessary to understand and correct what the AI produced, while the AI tool saved the time that would otherwise be spent on repetitive typing.

5.5 Automation Result

The n8n workflow performed the task it was designed for: every morning it reads the customer list, selects the customers whose payment is due within two days, sends them an e-mail through Gmail and marks the record. The task that would otherwise require a person to check dates and write messages was reduced to a one-time setup.

5.6 Internship Experience

The internship at QIQ AI was a practical introduction to professional software work. Beginning as a fresher, the intern was guided step by step by the mentor, from the first HTML page to complete websites. Working offline at the company gave direct exposure to a working environment, quick feedback on the code and the habit of finishing a task and presenting it. The completion certificate issued by the Founder confirms the successful completion of the internship.

5.7 Discussion

The results show that a beginner with sound fundamentals can build good-looking websites quickly by using an AI-assisted tool correctly. The most important lesson was that AI does not replace understanding: every website needed review, small corrections and testing before it was acceptable. The Python and automation modules showed that the same discipline of breaking a problem into steps applies to programs, to workflows and to websites.

The websites developed are front-end projects and use sample content. With a back-end, a database and real data, they could be developed further into complete products, as suggested in the next chapter.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion of Individual Modules

- **Web fundamentals:** the structure of web pages and the roles of HTML, CSS and JavaScript were understood and applied.
- **AI-assisted development:** Google Antigravity was used to plan and generate websites, with the generated code reviewed and corrected.
- **Websites:** Coop AI, Buhari and a Snapchat-style landing page were developed with responsive, themed designs.
- **Python and Git:** basic programs were written and a project was placed under version control.
- **Automation and marketing:** an n8n workflow sends payment-due notifications by Gmail, and the basics of online promotion were learned.

6.2 Overall Conclusion

The Software Developer Internship at QIQ AI gave the intern practical experience of the whole path from learning to building. Starting from the basics of web coding, the intern progressed to building complete websites with an AI-assisted tool, learned Python, Git and GitHub, automated a business task with n8n and understood how products are marketed online. The internship showed that modern development is a partnership between human understanding and AI assistance, and that careful review and testing remain the developer's responsibility.

6.3 Future Work

- Add a back-end with Node.js or Python (Flask) so that bookings and orders are stored.
- Connect a database such as Firebase or MySQL for users, services and orders.
- Add secure login and sign-up with user accounts.
- Integrate an online payment gateway for the Coop AI bookings and the Buhari orders.
- Deploy the websites with GitHub Pages, Netlify or Vercel on a custom domain.
- Extend the n8n workflow with SMS or WhatsApp alerts and a payment-received confirmation.

6.4 Skills Acquired

- **Technical:** HTML, CSS, JavaScript, Python basics, Git, GitHub and n8n.
- **AI skills:** prompt writing, reviewing AI-generated code and debugging.
- **Professional:** communication with a mentor, time management and documentation.

6.5 Limitations of the Present Work

- The websites are front-end only and do not yet have a real back-end or database.
- The content, such as services, dishes and prices, is sample content for learning.
- The internship was short (three weeks), so each topic was covered at a foundational level.

- Testing was manual; automated testing was not part of the internship.

6.6 Benefits of the Internship

For the intern, the internship provided hands-on skills, a portfolio of three websites and an automation workflow, and confidence in using modern AI tools responsibly. For an organisation, the approach shows how a small team can deliver working prototypes quickly and automate routine tasks, which saves time and effort.

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APPENDIX

INTERNSHIP COMPLETION CERTIFICATE



Date : 18.08.2026

Dear, Sakthivel

It is our pleasure to confirm that Sakthivel has successfully completed an internship at QIQ AI, from 10th May 2026 to 31st May 2026.

During this period, , Sakthivel held the position of Software developer Intern, contributing significantly to various projects and initiatives within our organization. Throughout the internship, Sakthivel exhibited exceptional creativity, diligence, and a strong aptitude for software development and delivered high-quality work. We are confident that the knowledge and experience , Sakthivel gained during his time with us will serve him well in his future endeavors. He has proven to be a valuable asset to our team.

We wish him all the best and good luck for his future endeavors and academic goals.

Sincerely,
For QIQ AI

Abinav Satish,
Founder

QIQ AI No 2/41, Masilamani Road, Balaji Nagar, Royapettah 600041

evolve@qiqai.in

Certificate of Completion issued by QIQ AI, Chennai (dated 18.08.2026)